Selectable Experts and Contextual Computation

An evidence-based argument about learning history and MoE reuse

Research argument draft — not a submission

29 October 2026

Abstract

What does learning to select expert subsets establish about the division of computation in a mixture-of-experts model? We distinguish three relationships: executing a useful subset inside a trained model, transporting a predictive property with a parameter package, and reusing that package's realized computation in another context. Across mature, architecture-matched public training paths, better selective execution coexists with stronger dependence on shared computation at similar intact quality. Source interventions identify routed functions that transport a low-budget likelihood advantage without transporting intact quality. On a fixed panel of 55 additional subjects, this advantage persists after clamping realized routes, while the removal cost of unchanged shared functions also increases. Thus, these effects cannot be explained solely by the routing changes induced by the transported functions. Cross-context replay of actual shared messages then reveals a prediction cost that largely survives local norm and projection correction. Together, the results characterize selectable computation through its learned production and use relationships, rather than through expert identity alone. The transported advantage concerns probability quality; an accuracy advantage remains unresolved.

1 The question: what becomes modular?

Sparse activation makes a large function bank computationally affordable, but does not by itself specify how that bank divides a task. Training can constrain expert selection, and selective execution can retain useful task performance [1]. The scientific question is what kind of computational organization this success identifies. Does a successful subset behave as an independently reusable collection of functions, or does its utility depend on relationships with the rest of the learned system?

This question is compatible with cooperation being useful for deployment. Interaction-aware pruning already exploits cooperative structure [2], and retaining shared computation can preserve substantial performance during condensation [3]. Our target is therefore not a generic claim that experts interact. We ask how learning history changes the relationship between *selection*, *property transport*, and *predictive dependence*, and use interventions to locate where these relationships are implemented.

The evidence supports three connected statements. First, improved selective execution can coexist with greater reliance on shared computation. Second, the parameter package carrying the selective-execution advantage need not be the partner whose native deletion is most damaging. Third, unchanged functions—and even their realized messages—can have different predictive roles in different computational contexts. The realized-route intervention connects the second and third statements by excluding a routing-only explanation within the controlled receiver.

2 Objects, interventions, and evidence

We study four public mature paths from the EMO release [1]: a standard pre-annealing parent P, standard annealing A, EMO annealing B, and full-course EMO F. These are architecture-matched public histories, not a verified population of independent initializations or an isolated timing treatment. Let D denote backbone, attention and readout, G the gate parameters, R the routed expert functions, and S the always-active shared functions. An R/S intervention covers the corresponding suites in all 16 layers; it is not evidence for a smallest single-expert unit.

For an intact loss L, we measure selection by the loss increase under a calibration-defined expert pool, alongside fixed random pools of the same size. We measure property transport by changing a parameter source while holding a common receiver fixed. We measure a function's predictive role by its removal effect in the specified system. These are operationally different questions, not alternative names for one modularity score.

Gate parameters held fixed do not imply realized routing held fixed: changed expert outputs can change later hidden states and gate outputs. We therefore capture actual gate logits before the common pool mask and replay them while preserving the original softmax, top-k, multiplication, and accumulation program. This clamps both support and weights. If z denotes a replayed trajectory,

the shared role in that condition is

$$\Delta_S(R, z) = L(D_P, G_P, R, 0; z) - L(D_P, G_P, R, S_P; z). \quad (1)$$

Other responses remain endogenous. These interventions do not identify an isolated expert direct effect or a unique mediation percentage.

Language-model measurements use 64 windows and 20 file clusters. The initial task panel contains 128 biology and mathematics questions. The route experiment adds a fixed 440-question panel: eight questions in each of the other 55 MMLU subjects, with separate validation calibration. The panels use different calibration sample counts, so between-panel magnitude differences cannot be assigned solely to task content. We report answer-token NLL, choice CE, accuracy, and output KL separately. Confidence intervals are paired input resampling intervals conditional on the models; they are not training-seed intervals.

3 Three results and their connection

3.1 Selection improves while shared dependence grows

The late- and full-course EMO endpoints B/F have intact text NLL 2.6218/2.6212 and identical task raw accuracy 39.06%. Yet F has lower selected-pool damage, while removing shared computation adds 1.8509 more text NLL damage than in B (95% CI $[1.3311, 2.3010]$). The corresponding task answer-NLL difference is 5.3112; the raw-accuracy damage difference remains unresolved. Random-pool controls show that general robustness and useful selection coexist. The result is a measured coexistence of selectability and cooperation, not an assertion that cooperation obstructs selection (Figure 1).

3.2 A property carrier also changes the role of an unchanged partner

In the common P receiver, replacing R_P by R_B transfers a low-budget answer-NLL advantage on the original panel, whereas replacing shared functions alone does not transfer the same property. Full-quality transport fails: the original-panel full NLL increases 0.0802 while low-budget NLL decreases 0.1882. Holding S_P fixed and changing R increases its removal effect by 0.3386. The function package that carries a useful operating property and the native necessary partner must therefore be identified separately.

The larger subject panel strengthens this statement and locates its runtime implementation (Figure 2). The natural low-budget NLL advantage is 0.4207 ($[0.3221, 0.5234]$). At fixed P/B routes, the routed-function source effect remains an improvement of 0.2937/0.3824, while the unchanged shared suite's removal cost increases 0.2891/0.1840; all four intervals exclude zero. Routing still matters: the B trajectory improves R_B by a further 0.1270, and the function–trajectory interaction is -0.0887 ($[-0.1242, -0.0500]$). The data support neither a routing-only explanation nor route-independent autonomous components.

The choice CE advantage is 1.1386 ($[0.8431, 1.4676]$), but the raw accuracy difference is +1.14 percentage points ($[-2.27, 4.55]$). Importantly, shared dependence here is measured in the full regime and the transported advantage in the low-budget regime. Their coexistence does not yet establish shared compensation as the cause of the budget advantage.

3.3 Realized messages retain context-dependent value

Function identity leaves production and use entangled. We therefore fix shared functions and cross their actual messages between routed-function contexts on the same input. Foreign messages have an average answer-NLL cost of 0.2605 relative to their original pairing. Correcting the consumer's clean norm and residual projection reduces this cost only to 0.2459 ($[0.2096, 0.2819]$). The improvement is 0.0146 ($[0.0050, 0.0250]$), concentrated in biology rather than mathematics; choice CE and accuracy do not show equivalent recovery (Figure 3). The tested local scalar interface does not account for most of the prediction mismatch. This is not a proof against richer learned alignment, nor does arbitrary direction destruction identify semantic content.

4 Position in the literature and next discriminating test

Expert specialization, cooperation, and reusable features have substantive precedents. EMO establishes a training route to selective deployment [1]; HOPE demonstrates a deployment benefit from retaining cooperative expert structure [2]; CD-MoE demonstrates effective condensation that retains shared computation [3]. Spectral decoupling changes parameter and gradient sharing and improves training [4]; its common matrices are not the same object as a replayed shared-expert message. Basic–refinement attribution already supplies real shared/routed ablations [5]. Stitching provides genuine functional-feature transport after alignment [6], while direct and invariance-aware matching clarify what successful alignment establishes [7, 8]. Our contribution is the combined identification of *which function package carries a deployment property*, *how an unchanged partner's role changes*, and *which realized routes and message interfaces account for these effects* in mature MoE histories.

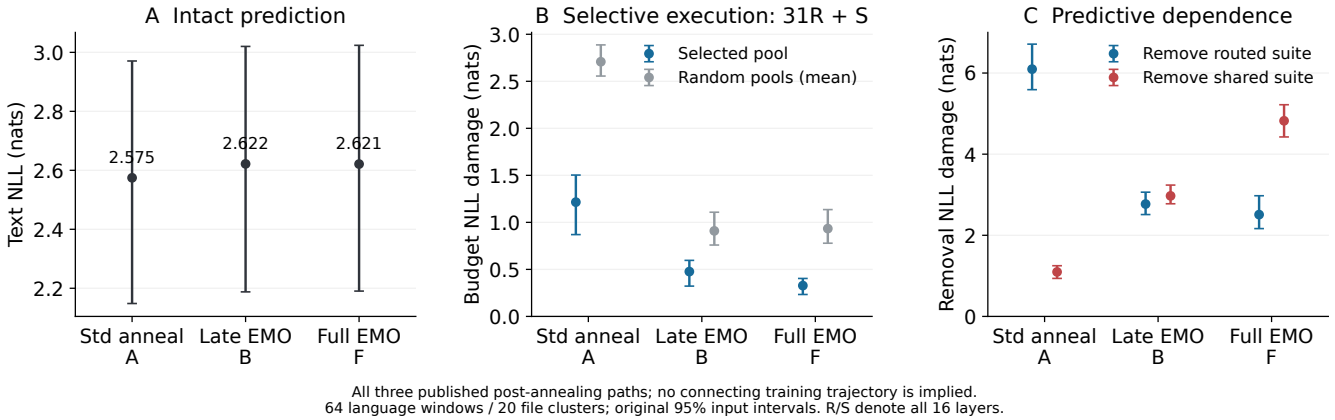


Figure 1: All three post-annealing paths, with intact quality, selected and random pool damage, and routed/shared removal effects separated. The groups are not connected as a training trajectory. Original paired file-cluster intervals are retained. Source estimates: E162/E163.

The appropriate scale is a causal study of these relationships in mature systems, not a universal theory of single-expert semantic identities. The next registered test uses a second mature public lineage: OLMoE's 5.033T checkpoint and its ordinary 100B annealing endpoint [9]. It compares intact and selected-pool quality with a full backbone–gate–expert source cross and a targeted realized-route cross. This architecture has no always-active shared expert, so the test does not invent a shared group to reproduce the original result. A compatible outcome is scientifically informative: it would weaken the hypothesis that any mature training change necessarily breaks component reuse. No result from this registered test is included in the present draft.

5 Traceability and remaining scope

All plots copy existing audited point estimates and original confidence intervals; their sidecar gives JSON paths and SHA256 for every value. The complete grids, reverse directions, accuracy results, failed attempts and numerical-source packages remain available. Identity replays, causal-prefix checks, source read-only checks and independent formula recomputation validate the measurement program; they do not substitute for an independent training population. The native formation trajectories, smallest reusable units, and the scope across mature lineages remain questions to investigate. This argument is anchored in observed separations and controlled pathways, without requiring them to be universal or treating a nearby method as a reason to abandon the scientific question.

References

[1] R. Wang et al. *EMO: Pretraining Mixture of Experts for Emergent Modularity*. 2026. https://arxiv.org/abs/2605.06663.

[2] A. M. Tseng et al. *Higher-order pruning of experts in mixture-of-experts language models*. arXiv:2609.18916v3, 2026.

[3] M. Cao et al. *Condense, Don't Just Prune*. TMLR, 2025; reading version arXiv:2412.00069v3.

[4] R. Huang et al. *SD-MoE: Spectral Decomposition for Effective Expert Specialization*. ICML, 2026.

[5] J. Li et al. *Decoding Knowledge Attribution in Mixture-of-Experts: A Framework of Basic-Refinement Collaboration and Efficiency Analysis*. ACL, 2025.

[6] *Transferring Linear Features Across Language Models With Model Stitching*. NeurIPS, 2025.

[7] *How Not to Stitch Representations to Measure Similarity*. AAAI, 2025.

[8] *Grounding Functional Similarity by Invariance-Aware Model Stitching*. ICML, 2026; reading version arXiv:2505.20142v2.

[9] N. Muennighoff et al. *OLMoE: Open Mixture-of-Experts Language Models*. ICLR, 2025. https://huggingface.co/allenai/OLMoE-1B-7B-0924.

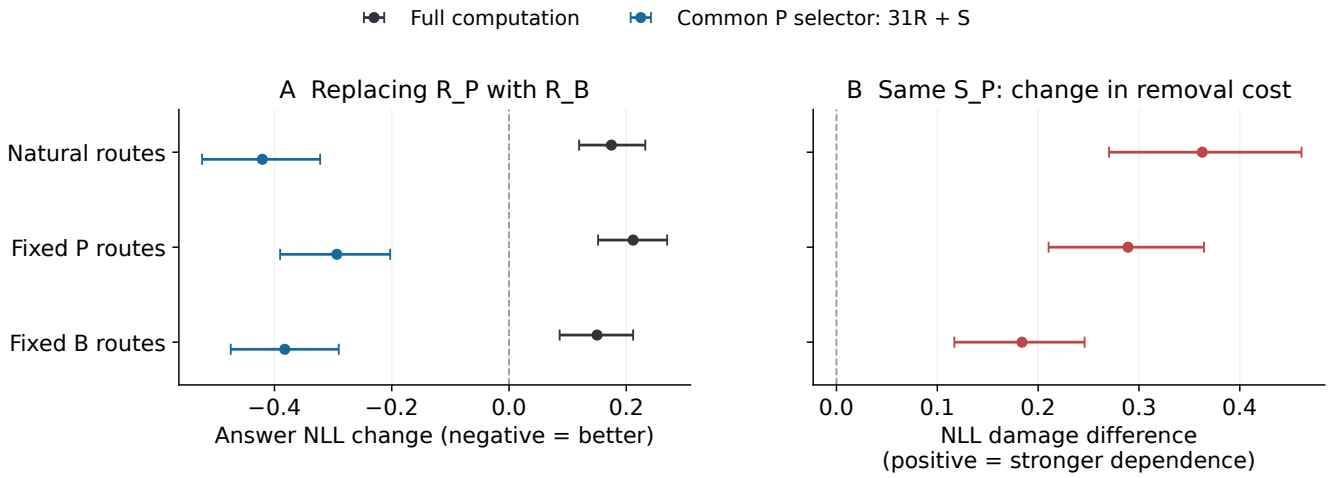


Figure 2: Two effects survive realized-route fixation on all 55 subjects. Left: changing only the routed function source improves low-budget likelihood while worsening full likelihood. Right: the same change increases the removal cost of unchanged shared functions. The actual support and weights are clamped; other response paths remain endogenous. Original subject-cluster intervals and accuracy boundary are shown. Source: E164.

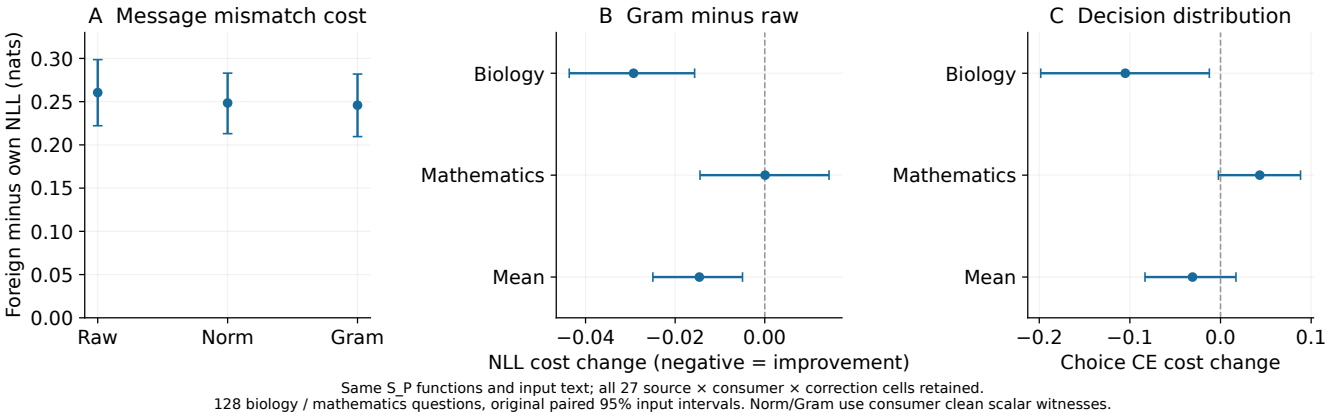


Figure 3: Actual-message reuse after local correction. All 27 source–consumer–correction cells are retained in the underlying results. Norm and Gram corrections give modest answer-NLL recovery; the subject and choice-CE panels retain the heterogeneous outcomes. Source: E163.